

**A SMART MUSHROOM INCUBATION SYSTEM FOR
ENHANCED YIELD AND SUSTAINABILITY IN SMALL-SCALE
FARMING**

**Fungal Contamination Detection Module using Mobile
Imaging and Deep Learning**

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ABSTRACT

Mushroom cultivation is an important component of sustainable agriculture due to its high nutritional value and low production cost. In Sri Lanka, oyster mushroom farming is widely practiced by small-scale farmers; however, it is significantly affected by fungal contamination, inconsistent yield cycles, and labor-intensive manual monitoring processes, leading to productivity losses [1], [2].

Traditional visual inspection methods are often subjective and fail to detect early-stage fungal infections such as green mold and cobweb disease, which can rapidly spread under suitable environmental conditions [3], [4]. This highlights the need for an automated and intelligent detection system that can support early intervention and reduce crop damage.

This research proposes a Fungal Contamination Detection Module using Mobile Imaging and Deep Learning, designed to identify fungal infections in mushroom cultivation environments. The system captures images of mushroom packets using a mobile-based imaging setup and applies a lightweight Convolutional Neural Network (CNN) for real-time classification of contamination types [5], [6].

Unlike traditional image processing methods, CNN-based models automatically extract hierarchical features from raw images, improving robustness and classification accuracy under varying environmental conditions [7], [8]. The system also integrates a real-time alert mechanism and a feedback-based learning module, allowing farmers to validate predictions and continuously improve model performance [9].

Experimental studies in controlled environments show that the proposed system achieves high detection accuracy while reducing manual inspection effort and improving early disease identification. These results demonstrate the effectiveness of deep learning and mobile imaging technologies in smart agriculture applications [10], [11], [12].

KEYWORDS

Mushroom Cultivation, Fungal Contamination Detection, Deep Learning, Convolutional Neural Network (CNN), Mobile Imaging, Smart Agriculture, Precision Farming, Image Classification, Real-Time Monitoring

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

Adam – Adaptive Moment Estimation

API – Application Programming Interface

BSc – Bachelor of Science

CNN – Convolutional Neural Network

CO₂ – Carbon Dioxide

CSS – Cascading Style Sheets

CV – Computer Vision

DL – Deep Learning

GPU – Graphics Processing Unit

HTML – HyperText Markup Language

IDE – Integrated Development Environment

IoT – Internet of Things

KNN – K-Nearest Neighbors

R-CNN – Region-Based Convolutional Neural Networks

RESTful – Representational State Transfer

RMSProp – Root Mean Square Propagation

SaaS – Software-as-a-Service

SDGs – Sustainable Development Goals (United Nations)

SGD – Stochastic Gradient Descent

SLIIT – Sri Lanka Institute of Information Technology

SVM – Support Vector Machines

YOLO – You Only Look Once

1 INTRODUCTION

Mushroom cultivation has emerged as a significant component of modern sustainable agriculture, particularly in developing countries such as Sri Lanka, where small-scale farming plays a vital role in rural livelihoods. Among different mushroom varieties, oyster mushrooms (*Pleurotus* spp.) are widely cultivated due to their relatively simple cultivation process, short production cycle, and high nutritional and economic value. They are rich in proteins, vitamins, and minerals, making them an important dietary supplement in both local and export markets. As a result, mushroom farming has increasingly become a source of income diversification for rural farmers, contributing to poverty reduction and food security in agricultural communities.

Despite its advantages, mushroom cultivation is highly sensitive to environmental conditions and requires strict monitoring of temperature, humidity, ventilation, and hygiene. Even minor deviations in these conditions can lead to the rapid development of fungal contamination, which is one of the most critical threats in mushroom production systems. Contamination can originate from several sources, including improperly sterilized substrates, contaminated spawn, airborne fungal spores, and poor handling practices during incubation and harvesting stages. Once contamination begins, it spreads quickly across cultivation bags or trays, often rendering entire batches unusable and causing substantial economic losses for farmers [8], [9].

In traditional mushroom farming practices, contamination detection is primarily carried out through manual inspection. Farmers or workers visually examine mushroom bags or growth environments to identify abnormal color changes, mold growth, or unusual texture variations. While this approach has been used for decades, it suffers from several critical limitations. Firstly, it is highly dependent on human experience and expertise, meaning that less experienced farmers may fail to detect early-stage contamination. Secondly, manual inspection is time-consuming and labor-intensive, especially in large-scale or semi-commercial farming environments. Thirdly, frequent physical handling of mushroom bags increases the risk of cross-contamination, as fungal spores may unintentionally spread from infected to healthy substrates during inspection.

Another major limitation of manual inspection is the delay in detection. Fungal contamination often begins at a microscopic level before becoming visually identifiable. By the time contamination is noticeable to the human eye, it may already have spread extensively within the cultivation system. This delay significantly reduces the chances of successful intervention and leads to reduced yield quality and economic loss. Therefore, there is a growing need for more efficient, automated, and intelligent systems capable of early-stage contamination detection in mushroom cultivation environments.

In recent years, advancements in Artificial Intelligence (AI), particularly in Deep Learning (DL) and Computer Vision (CV), have revolutionized the field of agricultural monitoring and disease detection. AI-based systems have demonstrated remarkable success in identifying plant diseases through image-based analysis, enabling early detection and automated decision-making. These

systems leverage large datasets of agricultural images and use deep neural networks to extract complex patterns that are often difficult for human observers to identify [10], [11].

Among deep learning techniques, Convolutional Neural Networks (CNNs) have become the most widely used architecture for image classification tasks in agriculture. CNNs are particularly effective because they automatically learn spatial hierarchies of features from input images, eliminating the need for manual feature extraction. Over the past decade, CNN-based models have achieved high accuracy in detecting diseases in various crops such as rice, potato, mango, and wheat. For example, deep learning models have been successfully applied to classify rice plant diseases with high precision, demonstrating the potential of AI in agricultural diagnostics [12], [13].

In addition to classification tasks, object detection frameworks such as You Only Look Once (YOLO) and Faster R-CNN have enabled real-time detection of multiple disease regions within a single image. These models are capable of identifying and localizing affected areas, making them highly suitable for field-level applications where multiple infections may occur simultaneously. Their ability to process images in real time has made them increasingly popular in smart agriculture systems [14], [15].

Parallel to advancements in deep learning, the integration of Internet of Things (IoT) technologies has further enhanced the capabilities of agricultural monitoring systems. IoT-based sensors enable continuous monitoring of environmental parameters such as humidity, temperature, and carbon dioxide levels, which are critical factors influencing mushroom growth. When combined with AI-based predictive models, IoT systems can provide real-time insights and automated alerts, improving decision-making efficiency in precision agriculture [8], [18].

Despite these technological advancements, most existing AI-driven agricultural systems are designed for large-scale commercial farming or controlled laboratory environments. These systems often rely on fixed imaging setups, high-performance computing infrastructure, or cloud-based processing, which limits their applicability in small-scale farming contexts such as rural mushroom cultivation in Sri Lanka. Furthermore, many systems are not designed for mobile deployment, making them inaccessible to farmers who rely on handheld devices for farm management.

Another limitation of existing systems is the lack of adaptability to real-world farming conditions. Environmental variability, inconsistent lighting conditions, and diverse cultivation setups often reduce the accuracy of AI models when deployed outside controlled environments. Additionally, most systems do not incorporate feedback mechanisms that allow continuous model improvement based on user input or new data. This results in static models that may degrade in performance over time when exposed to new conditions.

Therefore, there is a clear need for a more practical, mobile-friendly, and adaptive intelligent system specifically designed for mushroom cultivation environments. Such a system should be

capable of capturing real-time images using mobile devices, analyzing them using deep learning models, and providing immediate feedback to farmers in an accessible and user-friendly manner. It should also be cost-effective to ensure adoption among small-scale farmers who may not have access to advanced agricultural technologies.

This research aims to address these challenges by developing a mobile-based intelligent system for early detection of fungal contamination in mushroom cultivation. The proposed system integrates deep learning-based image analysis with mobile computing and real-time alert mechanisms to provide an efficient solution for farmers. By leveraging CNN-based models and mobile imaging capabilities, the system aims to improve early detection accuracy while reducing dependency on manual inspection methods.

Furthermore, the system is designed to incorporate a feedback-based learning mechanism that allows continuous improvement of the model over time. This ensures that the system remains adaptive to different environmental conditions and cultivation practices. By focusing on affordability, scalability, and usability, the proposed solution aims to bridge the gap between advanced AI technologies and practical agricultural applications in rural Sri Lanka.

Overall, this research contributes to the field of smart agriculture by introducing a context-aware, mobile-based deep learning system tailored for mushroom cultivation. It aligns with global efforts toward sustainable agriculture and digital transformation in farming practices, as emphasized in the United Nations Sustainable Development Goals (SDGs), particularly those related to zero hunger, responsible production, and sustainable economic growth [20].

1.1 Background and Literature Survey

Mushroom farming plays a vital role in Sri Lanka's agricultural economy, especially among rural farming communities where it serves as a low-cost, high-return agricultural activity. Oyster mushrooms are particularly popular due to their adaptability to local climatic conditions and relatively simple cultivation requirements. However, despite these advantages, mushroom cultivation is highly vulnerable to fungal contamination, which significantly affects productivity and profitability.

Fungal contamination typically occurs due to environmental imbalances, poor sterilization techniques, and improper handling during cultivation. Once contamination spreads, it can rapidly destroy entire batches of mushroom bags, leading to substantial financial losses. Studies have shown that early detection of contamination is critical in minimizing crop damage and improving yield efficiency [8], [9].

Traditional approaches to contamination detection rely on manual inspection, which is inefficient and prone to human error. Manual methods are unable to detect early-stage contamination accurately, as fungal growth often begins at a microscopic level. Additionally, these methods are

labor-intensive and not suitable for large-scale monitoring. This has led to increasing interest in automated detection systems powered by AI and machine learning technologies.

Recent advancements in AI have enabled the development of intelligent agricultural systems capable of analyzing plant health through image processing techniques. Deep learning models, particularly CNNs, have been widely used for plant disease detection due to their high accuracy and ability to learn complex patterns from image data [10], [11].

Several studies have demonstrated the effectiveness of CNN-based models in agricultural disease classification. For instance, rice plant disease detection using CNN architectures has achieved high accuracy, highlighting the potential of deep learning in agricultural diagnostics [12]. Similarly, mobile-based deep learning models have been evaluated for plant disease surveillance, showing promising results in real-world applications [13].

Object detection frameworks such as YOLO and Faster R-CNN have further enhanced agricultural monitoring by enabling real-time detection and localization of disease regions within images [14], [15]. These models are particularly useful in dynamic environments where multiple infections may occur simultaneously.

In addition to deep learning advancements, IoT-based agricultural systems have gained attention for their ability to monitor environmental conditions in real time. IoT sensors provide continuous data on temperature, humidity, and other critical factors influencing crop growth. When integrated with AI models, these systems enable predictive analytics and automated decision-making [8], [18].

However, despite these advancements, most existing systems are not designed for small-scale mushroom farming environments. They often require fixed infrastructure, high computational resources, and stable internet connectivity, making them unsuitable for rural deployment. Furthermore, there is a lack of mobile-based solutions that allow farmers to perform real-time contamination detection using handheld devices.

Existing systems also lack adaptive learning capabilities, where models improve over time based on new data or user feedback. This limitation reduces their long-term effectiveness in dynamic agricultural environments. Additionally, affordability remains a major barrier, as many AI-based agricultural systems are expensive to implement and maintain.

Therefore, there is a clear research gap in developing a mobile-based, cost-effective, and adaptive AI system specifically tailored for mushroom cultivation. Such a system should integrate image capture, real-time analysis, and feedback-based learning to provide a comprehensive solution for farmers.

1.2 Research Gap

Despite significant advancements in Artificial Intelligence (AI), Deep Learning (DL), and Internet of Things (IoT) technologies in the agricultural domain, there remains a clear gap between existing research outputs and practical real-world deployment, particularly in small-scale mushroom cultivation systems.

Most current AI-based agricultural disease detection systems are designed primarily for large-scale crop monitoring or controlled laboratory environments. These systems typically rely on high-resolution imaging setups, fixed camera infrastructures, or cloud-based processing pipelines that require stable internet connectivity and significant computational resources. While these approaches have demonstrated high accuracy in plant disease detection tasks, they are not easily transferable to small-scale farming contexts, especially in rural areas where infrastructure and technological access are limited [16], [17].

Another key limitation in existing research is the lack of mobile-first agricultural disease detection systems. Although mobile-based plant disease detection models have been explored in some studies, they are generally limited to offline classification tasks without real-time monitoring capabilities or integration with farm-level decision-making systems. This restricts their usability in dynamic agricultural environments where immediate detection and response are critical.

Furthermore, most existing systems focus on general crop diseases such as rice, potato, or leaf-based plant infections, rather than domain-specific applications such as mushroom cultivation. Mushroom farming presents a unique environmental and biological context, where contamination often occurs in enclosed incubation conditions and spreads rapidly across substrate packets. However, very limited research has been conducted on applying AI-based contamination detection specifically to mushroom cultivation environments.

Another significant gap is the absence of adaptive learning mechanisms in most agricultural AI systems. Existing models are typically trained once on static datasets and deployed without continuous improvement. In real-world farming conditions, environmental variability such as lighting changes, humidity fluctuations, and different cultivation setups can significantly affect model performance over time. The lack of feedback-driven learning systems prevents these models from adapting to new conditions or improving based on farmer interactions.

Additionally, affordability and accessibility remain major concerns. Many existing AI and IoT-based agricultural solutions require expensive sensors, high-performance GPUs, or cloud computing services, making them unsuitable for small-scale farmers in developing countries such as Sri Lanka. There is a clear need for lightweight, cost-effective, and mobile-compatible solutions that can be widely adopted without requiring advanced technical infrastructure.

Most importantly, there is currently no fully integrated system that combines mobile-based image acquisition, CNN-based real-time contamination detection, instant alert mechanisms, and

feedback-based model improvement specifically tailored for mushroom cultivation environments. This lack of integration represents a significant research gap that this study aims to address.

1.3 Research Problem

Mushroom cultivation has become an increasingly important agro-based industry in Sri Lanka, particularly among small-scale rural farmers due to its low initial investment, short production cycle, and high nutritional and commercial value. However, despite its economic potential, the industry faces a persistent and critical challenge: fungal contamination during the incubation and cultivation stages, which directly affects yield quality, productivity, and profitability.

Contamination in mushroom cultivation environments can spread rapidly across substrate bags, often resulting in complete batch failure if not detected at an early stage.

Currently, most mushroom farmers rely on manual visual inspection to identify contamination. This traditional approach depends heavily on human expertise, experience, and observational accuracy. However, manual inspection is inherently limited in detecting early-stage fungal infections, which often appear at a microscopic or barely visible level before becoming fully developed. As a result, contamination is frequently identified only after it has already spread significantly, leading to irreversible crop loss. Moreover, manual inspection introduces inconsistencies in detection due to subjective judgment and fatigue, especially when monitoring large numbers of cultivation packets over time.

Another significant issue associated with manual detection is the risk of cross-contamination. During physical inspection, farmers may unintentionally transfer fungal spores from infected to healthy mushroom bags, thereby accelerating the spread of contamination within the cultivation environment. This not only increases economic losses but also reduces overall production efficiency and farm sustainability.

Recent research in agricultural disease detection highlights the effectiveness of Artificial Intelligence (AI), Deep Learning (DL), and Computer Vision (CV) techniques in identifying plant diseases with high accuracy. Convolutional Neural Networks (CNNs), in particular, have demonstrated strong performance in image-based classification tasks due to their ability to automatically extract hierarchical visual features from agricultural datasets [13], [14], [15]. Furthermore, advanced object detection models such as YOLO and Faster R-CNN have enabled real-time detection of multiple disease regions within a single image, making them suitable for precision agriculture applications [33], [34], [36].

Additionally, the integration of Internet of Things (IoT) technologies in agriculture has enabled continuous environmental monitoring, allowing farmers to track critical parameters such as temperature, humidity, and CO₂ levels in real time. These environmental factors are directly linked to fungal growth and contamination risks in mushroom cultivation systems. IoT-based

smart agriculture systems have therefore been widely proposed as a solution for improving farm monitoring and decision-making processes [18], [19], [49], [50].

Despite these technological advancements, most existing AI-based agricultural systems are primarily designed for general crop disease detection, such as rice, potato, or leaf-based plant diseases, and are not specifically tailored for mushroom cultivation environments. Studies focusing on plant disease recognition using deep learning have shown strong performance in controlled datasets, but their applicability in real-world, small-scale farming environments remains limited [9], [17], [38]. Similarly, many mobile-based plant disease detection systems provide only offline classification capabilities and lack real-time monitoring, alert generation, or integration into farm management workflows [5], [16].

Another major limitation in existing solutions is their reliance on controlled imaging environments or fixed camera setups, which are not practical for small-scale mushroom farms. Mushroom cultivation typically takes place in enclosed or semi-controlled environments such as sheds or incubation rooms, where lighting conditions, humidity levels, and spatial arrangements vary significantly. These variations reduce the robustness of many existing AI models when deployed outside laboratory conditions [11], [14], [17].

Furthermore, most current systems do not support real-time decision-making or farmer-centric usability features. Even when AI models achieve high accuracy in disease detection, the lack of user-friendly interfaces, mobile integration, and real-time alert mechanisms limits their practical adoption among farmers. Studies have also identified a gap in integrating feedback-driven learning mechanisms, where models continuously improve based on real-world usage data and farmer input. Without such adaptability, AI models remain static and gradually lose effectiveness when exposed to new environmental conditions or unseen contamination patterns.

From an economic perspective, many existing AI and IoT-based agricultural solutions require expensive hardware components, high-performance computing infrastructure, or cloud-based processing services. These requirements make them unsuitable for deployment in rural and resource-constrained farming communities in developing countries such as Sri Lanka. This affordability barrier significantly restricts the adoption of advanced agricultural technologies among small-scale farmers [18], [19], [50].

Moreover, although significant research has been conducted on deep learning architectures such as MobileNet, EfficientNet, and ResNet for plant disease detection [27], [28], [37], most of these studies focus on classification accuracy improvements rather than end-to-end practical deployment in real agricultural workflows. There is still a lack of integrated systems that combine mobile image capture, real-time CNN inference, contamination alerting, and adaptive learning mechanisms specifically for mushroom farming environments.

Based on the reviewed literature, it is evident that there is a critical gap between advanced AI research and its real-world agricultural application, particularly in mushroom cultivation

systems. While AI models demonstrate strong theoretical performance, their practical usability in small-scale farming environments remains limited due to issues related to mobility, cost, adaptability, and real-time functionality.

1.4 Research Objectives

The primary objective of this research is to develop an intelligent, mobile-based deep learning system capable of detecting early-stage fungal contamination in mushroom cultivation environments in real time. The study is motivated by the need to improve the efficiency, accuracy, and accessibility of contamination detection in small-scale mushroom farming systems in Sri Lanka. By integrating Artificial Intelligence (AI), specifically Convolutional Neural Networks (CNNs), with mobile computing technologies, the proposed system aims to provide a practical and scalable solution that supports farmers in making timely and informed decisions, thereby reducing crop losses and improving overall productivity [20].

The main objective of this research is to design and implement a mobile-based system that enables real-time detection of fungal contamination in mushroom cultivation using deep learning techniques. Unlike traditional manual inspection methods, which are time-consuming, subjective, and prone to human error, the proposed system aims to automate the detection process using image-based analysis. This system is intended to be deployed on low-cost mobile devices, making it accessible to small-scale farmers who may not have access to advanced agricultural technologies or high-performance computing resources. The focus is on developing a light yet accurate solution that can operate effectively in real-world farming environments.

To achieve this, one of the key specific objectives of the study is to design and develop a mobile-based image acquisition system tailored for mushroom cultivation environments. This system will allow farmers to capture images of mushroom cultivation packets in a structured manner, such as row-by-row or batch-wise scanning. The objective is to ensure consistent image quality and coverage across different cultivation setups, enabling reliable input data for the deep learning model. This first mobile approach ensures that the system remains practical and user-friendly, even in resource-constrained rural environments where fixed imaging infrastructure is not available.

Another important objective is to develop a Convolutional Neural Network (CNN)-based deep learning model capable of accurately classifying mushroom cultivation images into healthy and contaminated categories. CNN architectures have been widely recognized for their effectiveness in image classification tasks due to their ability to automatically learn spatial and hierarchical features from raw image data [13], [14], [15]. In this research, the model will be trained using a curated dataset of mushroom cultivation images under varying environmental conditions to ensure robustness and generalization. The goal is to achieve high classification accuracy while maintaining computational efficiency suitable for mobile deployment.

A further objective of this research is to enable real-time contamination detection and alert generation. The proposed system will integrate the trained CNN model into a real-time inference pipeline that processes images captured through the mobile application. When fungal contamination is detected, the system will immediately generate alerts or notifications to inform farmers. This real-time capability is critical in minimizing the spread of contamination, as early intervention plays a key role in preventing large-scale crop damage. The system aims to bridge the gap between AI-based detection and practical farm-level decision-making.

In addition, the research aims to incorporate a feedback-based learning mechanism to improve model performance over time. This objective is based on the recognition that agricultural environments are dynamic, with variations in lighting, humidity, and cultivation conditions affecting model accuracy. By allowing farmers to provide feedback on predictions and by incorporating newly collected data into periodic model retraining, the system will be able to adapt and improve continuously. This adaptive learning approach ensures long-term reliability and enhances the system's applicability across different farming conditions.

Finally, the research aims to ensure that the proposed system is cost-effective, scalable, and suitable for deployment in small-scale farming environments. Many existing AI-based agricultural solutions require expensive hardware, cloud computing services, or complex infrastructure, which limits their adoption in developing regions. Therefore, this study focuses on developing a lightweight architecture that can operate efficiently on standard mobile devices without requiring high computational resources. The objective is to bridge the technological gap between advanced AI research and practical agricultural applications, making intelligent disease detection accessible to rural farmers in Sri Lanka and similar contexts.

Overall, these research objectives collectively aim to develop a comprehensive, mobile-based, and AI-driven solution that enhances early detection of fungal contamination in mushroom cultivation. The system is designed to improve agricultural productivity, reduce economic losses, and support sustainable farming practices through the integration of modern deep learning and mobile technologies.

2. METHODOLOGY

2.1 Methodology

The proposed fungal contamination detection system is designed as an integrated intelligent framework that combines deep learning, computer vision, mobile computing, and Internet of Things (IoT) technologies to address the limitations of traditional mushroom cultivation monitoring methods. In modern agricultural research, these technologies have become fundamental components of precision agriculture systems, enabling automated disease detection, environmental monitoring, and real-time decision support [1], [2]. The methodology adopted in this research focuses on building a practical, scalable, and mobile-first AI system specifically tailored for small-scale mushroom farming environments in Sri Lanka, where resource constraints and lack of continuous expert supervision often lead to delayed detection of fungal contamination and subsequent crop losses.

Unlike conventional agricultural AI systems that rely on controlled laboratory datasets or fixed imaging infrastructure, this study emphasizes real-world deployment conditions, where variability in lighting, humidity, background conditions, and cultivation structures significantly impacts system performance. Therefore, the proposed methodology is structured to ensure robustness, adaptability, and usability in real farming environments. Special attention is given to handling noisy and inconsistent image data captured through mobile devices, as well as integrating sensor-based environmental readings to improve the reliability of predictions under dynamic cultivation conditions.

Furthermore, the system is designed with a multi-layer architecture that supports continuous data acquisition, preprocessing, feature extraction, and classification using convolutional neural networks (CNNs). The integration of IoT sensors enables real-time monitoring of critical environmental parameters such as temperature, humidity, and CO₂ levels, which are strongly correlated with fungal growth patterns. These multimodal inputs enhance the decision-making capability of the system by combining visual symptom analysis with environmental context, thereby improving early detection accuracy and reducing false positives.

In addition, the methodology prioritizes accessibility and ease of use for farmers with limited technical expertise. A mobile-based interface is incorporated to allow users to capture images, receive instant feedback, and monitor cultivation conditions through a simple dashboard. This ensures that the system is not only technically robust but also practically deployable in rural and semi-urban agricultural settings. Overall, the proposed methodology aims to bridge the gap between advanced AI-driven agricultural research and real-world farming applications, contributing to improved productivity, reduced crop loss, and more sustainable mushroom cultivation practices.

2.1.1 Deep Learning and Computer Vision Foundation

Deep learning has revolutionized the field of agricultural image analysis by enabling systems to automatically learn meaningful feature representations directly from raw image data. Unlike traditional machine learning approaches, which require manual feature extraction such as color histograms, texture descriptors, or shape-based features, deep learning models eliminate this dependency by learning hierarchical representations through multiple layers of abstraction [3], [4].

In detection of agricultural diseases, computer vision combined with deep learning has shown exceptional performance in identifying complex patterns associated with plant infections, fungal growth, and leaf abnormalities. CNN-based architectures are particularly effective because they are capable of capturing spatial dependencies within images, which are critical for distinguishing between healthy and diseased samples [5], [6].

Extensive research has demonstrated the success of deep learning models in detecting diseases across a wide range of crops, including rice, potato, tomato, and mango leaves, under varying environmental conditions [3], [7]. These models consistently outperform traditional image processing techniques, particularly in scenarios involving noisy backgrounds and inconsistent lighting conditions, which are common in agricultural environments. This strong evidence supports the adoption of CNN-based deep learning as the core computational foundation of the proposed mushroom contamination detection system.

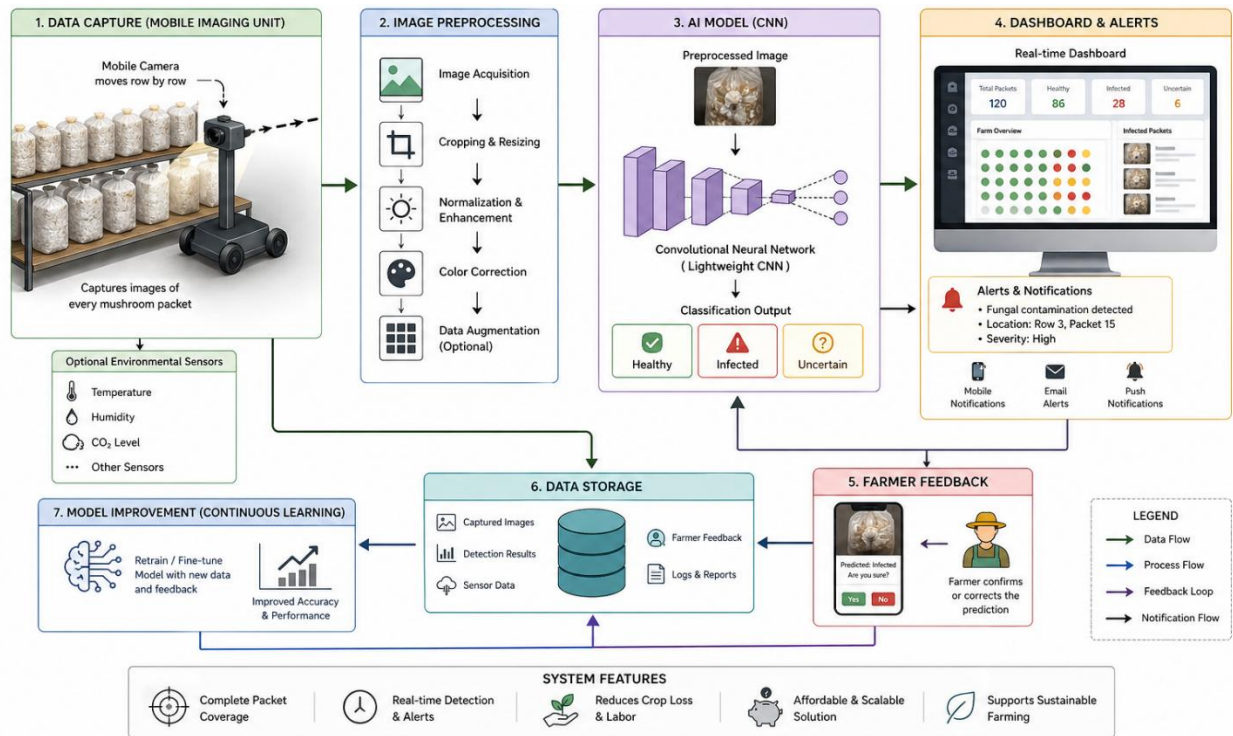


Figure 2.1.1 system diagram of the fungal contamination detection system

2.1.2 CNN-based Image Classification for Disease Detection

Convolutional Neural Networks (CNNs) form the backbone of modern image-based classification systems due to their ability to automatically extract hierarchical and spatial features from input images. A CNN typically consists of multiple layers, including convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification [8], [9].

In the context of agricultural disease detection, CNNs are particularly effective in identifying subtle variations in texture, color, and structural patterns that indicate infection or contamination. These variations are often too complex for manual feature engineering methods to capture effectively. CNNs learn these features directly from data, making them highly adaptable to different types of agricultural problems [10], [11].

Compared to traditional machine learning algorithms such as Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN), CNN-based systems demonstrate significantly higher accuracy and robustness, especially when dealing with large-scale image datasets [12]. Furthermore, CNNs are capable of generalizing well across different environmental conditions, provided that sufficient data augmentation and training strategies are applied.

Transfer learning plays a crucial role in improving CNN performance, particularly when labeled agricultural datasets are limited. By leveraging pre-trained models such as those trained on ImageNet, the system can inherit learned features and fine-tune them for mushroom contamination detection tasks, significantly reducing training time while improving accuracy [14], [15].

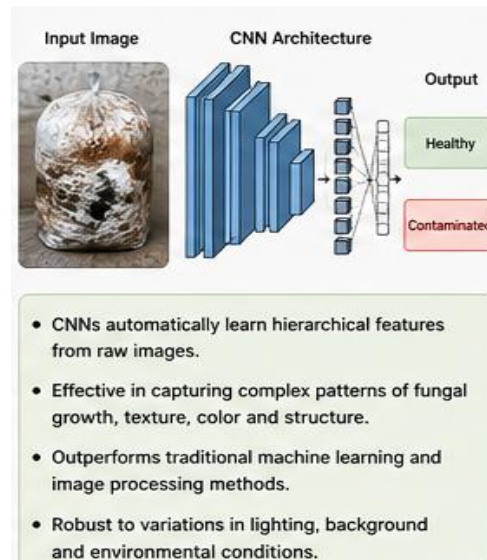


Figure 2.1.2.1 deep learning & computer vision foundation

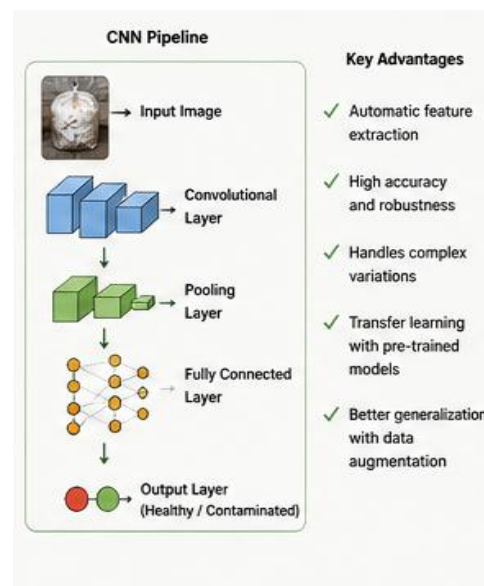


figure 2.1.2.2 cnn-based image classification for disease detection

2.1.3 Mobile-Based Disease Detection Systems

The increasing availability of smartphones and mobile devices has led to the emergence of mobile-based AI systems for agricultural monitoring. These systems enable farmers to perform real-time disease detection using handheld devices without requiring specialized imaging equipment or technical expertise [5], [16].

Mobile-based deep learning systems are particularly advantageous in rural agricultural settings where infrastructure limitations restrict access to advanced computing resources. Lightweight CNN architectures such as MobileNet and EfficientNet have been specifically designed for such environments, offering a balance between computational efficiency and predictive accuracy [17], [18].

These architectures utilize optimized convolution techniques such as depthwise separable convolutions, which significantly reduce the number of parameters and computational cost while maintaining high performance. This makes them suitable for deployment on low-end mobile devices commonly used by small-scale farmers.

In mushroom cultivation environments, mobile-based systems enable continuous monitoring by allowing farmers to scan cultivation packets directly using smartphone cameras. This eliminates the need for fixed camera installations and provides a flexible, user-friendly approach to contamination detection.

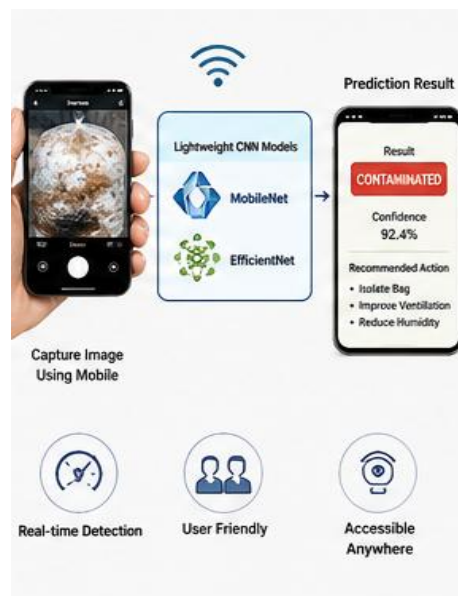


figure 2.1.3 mobile-based disease detection

2.1.4 IOT Integration in Agricultural Monitoring

The integration of Internet of Things (IoT) technologies into agricultural systems has significantly enhanced the ability to monitor environmental conditions in real time. IoT sensors can continuously collect data related to temperature, humidity, carbon dioxide levels, and air circulation, all of which are critical factors influencing mushroom growth and contamination risk [19], [20].

In mushroom cultivation systems, maintaining optimal environmental conditions is essential for preventing fungal contamination. Deviations in humidity or temperature can create favorable conditions for unwanted fungal growth. IoT-based monitoring systems provide real-time visibility into these environmental parameters, allowing farmers to take corrective actions before contamination spreads.

When combined with AI-based image analysis systems, IoT data enhances predictive accuracy and enables multi-modal decision-making. This hybrid approach allows the system not only to detect contamination visually but also to correlate environmental conditions with disease occurrence patterns [22], [23].

Such integration supports precision agriculture principles, where data-driven insights are used to optimize farming practices, reduce waste, and improve overall productivity.



figure 2.1.4 iot integration in agricultural monitoring

2.1.5 Data Acquisition and Preprocessing

Data acquisition is a critical step in the development of any deep learning-based agricultural system. In this study, image data is collected directly from mushroom cultivation environments under real-world conditions to ensure dataset diversity and robustness. Unlike laboratory datasets, real-world images capture variations in lighting, camera angles, background noise, and different stages of mushroom growth, which are essential for training a generalizable model [24], [25].

Although public datasets such as PlantVillage are widely used in agricultural research, they often lack environmental complexity and do not fully represent real farming conditions. Therefore, real-time image collection from actual cultivation environments plays a vital role in improving model reliability and performance [26].

Before training, all images undergo a comprehensive preprocessing pipeline. This includes resizing images to a uniform dimension, normalizing pixel values, and removing noise to standardize input data. Additionally, data augmentation techniques such as rotation, flipping, scaling, cropping, and brightness adjustment are applied to artificially expand the dataset [27], [28].

These augmentation strategies simulate real-world variations and help prevent overfitting by exposing the model to a wider range of possible input conditions. This significantly improves the generalization capability of the CNN model in real deployment scenarios [29].

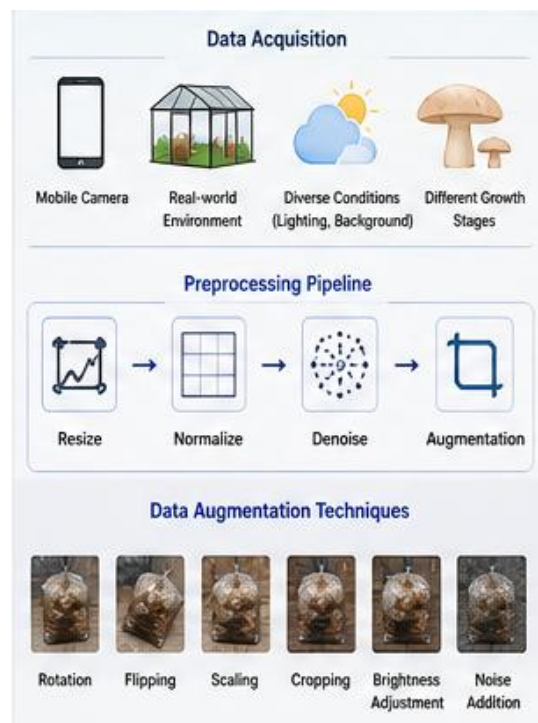


figure 2.1.5 data acquisition & preprocessing

2.1.6 Deep Learning Architectures and Optimization

The selection of appropriate CNN architecture is crucial for achieving optimal performance in fungal contamination detection. This study considers several state-of-the-art architectures, including MobileNet, ResNet, EfficientNet, and Xception, each offering unique advantages in terms of accuracy, efficiency, and scalability [30], [31].

MobileNet is particularly suitable for mobile deployment due to its lightweight structure and reduced computational complexity. It achieves efficiency through depthwise separable convolutions, making it ideal for real-time inference on mobile devices. ResNet introduces residual connections that help mitigate the vanishing gradient problem, enabling the training of very deep networks without performance degradation [30].

EfficientNet improves model scaling by uniformly balancing network depth, width, and resolution, achieving high accuracy with fewer parameters compared to traditional CNN architectures [37]. Xception further enhances feature extraction by using depthwise separable convolutions in an optimized architecture [29].

These architectures collectively provide a strong foundation for developing a robust and efficient contamination detection system suitable for real-world agricultural deployment.

2.1.7 Object Detection Techniques

While classification models can determine whether an image is healthy or contaminated, object detection models provide a more advanced level of analysis by identifying both the presence and precise spatial location of contamination within an image. This capability is particularly valuable in mushroom cultivation environments, where multiple contamination regions may occur simultaneously within a single cultivation bag or substrate block. By localizing infected areas, object detection enables targeted intervention, reduces unnecessary disposal of healthy sections, and supports more efficient farm management practices.

Object detection techniques generally involve predicting bounding boxes along with class labels for objects of interest within an image. Traditional detection approaches, such as Region-Based Convolutional Neural Networks (R-CNN) and its variants, operate in a two-stage pipeline where region proposals are first generated and then classified. Although these methods offer high accuracy, they are computationally intensive and less suitable for real-time applications in resource-constrained environments.

In contrast, single-stage detectors such as YOLO (You Only Look Once) have gained significant popularity due to their ability to perform real-time object detection with high efficiency [34]. YOLO treats object detection as a regression problem and processes the entire image in a single forward pass through the network. This unified architecture allows it to simultaneously predict

bounding box coordinates and class probabilities, significantly reducing computational overhead and enabling deployment on mobile and edge devices.

Subsequent versions of YOLO, including YOLOv3 and YOLOv4, have introduced several architectural enhancements such as multi-scale feature extraction, improved backbone networks (e.g., Darknet-53), and advanced training strategies [35], [36]. These improvements have resulted in better detection accuracy, especially for small and densely packed objects, which are common in agricultural scenarios like mushroom cultivation. Furthermore, the ability of YOLO models to balance speed and accuracy makes them particularly suitable for real-time monitoring systems, where immediate detection and response are critical to preventing the spread of fungal contamination.

In this research, object detection techniques are considered as a complementary extension to classification models, enabling not only the identification of contamination but also its localization, thereby enhancing the overall effectiveness of the system.

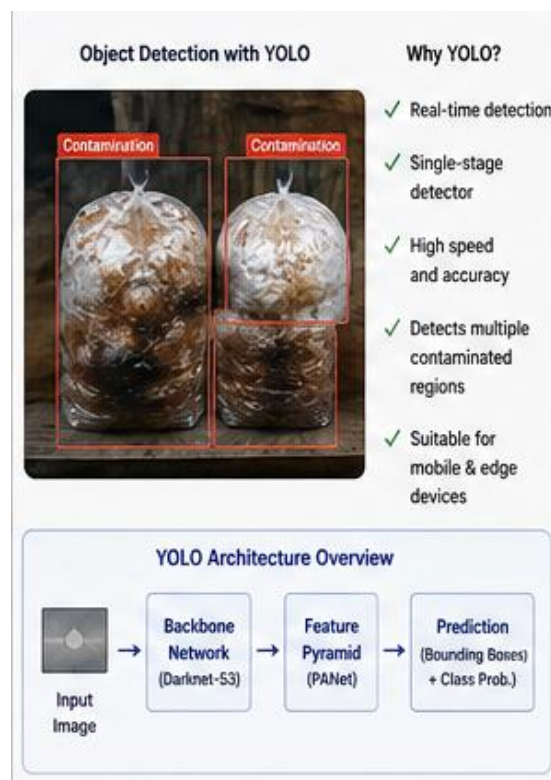


figure 2.1.7 object detection techniques

2.1.8 Machine Learning Optimization and Training

The successful training of deep learning models depends heavily on the selection of appropriate optimization algorithms and training strategies that ensure efficient convergence and high predictive performance. Optimization techniques are responsible for minimizing the loss function by iteratively updating model parameters based on the gradient of the error.

Stochastic Gradient Descent (SGD) remains one of the most widely used optimization algorithms due to its simplicity and effectiveness in handling large-scale datasets. However, its performance can be sensitive to the choice of learning rate and may converge slowly in complex optimization landscapes. To address these limitations, adaptive optimization algorithms such as Adam (Adaptive Moment Estimation) have been introduced [44], [45]. Adam dynamically adjusts the learning rate for each parameter by combining the advantages of momentum and RMSProp, resulting in faster convergence and improved stability during training.

The choice of loss functions also plays a critical role in model performance. For classification tasks, cross-entropy loss is commonly used as it quantifies the difference between predicted probability distributions and true class labels. Minimizing this loss enables the model to improve its prediction accuracy over time. In addition to the loss function, several regularization and normalization techniques are employed to enhance model generalization and prevent overfitting.

Batch normalization is used to stabilize and accelerate the training process by normalizing the input distributions of each layer, thereby reducing internal covariate shift. Dropout regularization randomly deactivates a subset of neurons during training, forcing the model to learn more robust and generalized features. Learning rate scheduling further improves optimization by dynamically adjusting the learning rate during training, allowing the model to make large updates initially and fine-tune parameters in later stages.

Transfer learning is another key strategy utilized in this research to overcome the limitations of small and domain-specific datasets. By leveraging pre-trained models trained on large-scale datasets such as ImageNet, the system can reuse learned feature representations and adapt them to the mushroom contamination detection task [46], [31]. This approach significantly reduces training time, improves accuracy, and enhances the model's ability to generalize to unseen data.

2.1.9 IOT-Based Smart Agriculture Systems

The integration of Internet of Things (IoT) technologies into agriculture has revolutionized traditional farming practices by enabling real-time monitoring, data-driven decision-making, and automation. IoT-based smart agriculture systems consist of interconnected sensors, communication networks, and cloud-based platforms that continuously collect and analyze environmental data [49], [50].

In the context of mushroom cultivation, environmental factors such as temperature, humidity, carbon dioxide levels, and light intensity play a critical role in determining growth conditions and susceptibility to fungal contamination. IoT devices equipped with environmental sensors can continuously monitor these parameters and transmit data to a centralized system for analysis. This real-time data acquisition enables early detection of unfavorable conditions that may promote contamination.

When integrated with AI-based image analysis systems, IoT provides valuable contextual information that enhances the accuracy and reliability of predictions. For instance, a sudden increase in humidity combined with visual signs of contamination can strengthen confidence of the detection model and trigger timely alerts. This multimodal approach improves the system's predictive capabilities and enables proactive intervention.

Moreover, IoT-based systems support remote monitoring and control, allowing farmers to access real-time data and receive alerts through mobile applications. This is particularly beneficial in rural and resource-limited settings, where continuous manual monitoring is not feasible. By enabling precise control over environmental conditions and early detection of risks, IoT systems contribute to improved productivity, reduced resource wastage, and enhanced sustainability in agricultural practices [20].

2.1.10 System Implementation Strategy

The proposed system implementation follows a unified and modular architecture that integrates mobile-based image acquisition, deep learning-based analysis, IoT-driven environmental monitoring, and real-time alert generation into a seamless workflow. The design prioritizes efficiency, scalability, and usability to ensure practical deployment in real-world mushroom farming environments.

The process begins with image acquisition, where farmers use a mobile device to capture images of mushroom cultivation packets. These images are then transmitted to the system and undergo preprocessing steps such as resizing, normalization, and noise reduction to ensure consistency and improve model performance. Data augmentation techniques may also be applied to enhance model robustness against variations in lighting, orientation, and background conditions.

Following preprocessing, the images are passed through a trained convolutional neural network (CNN) model for classification. The model analyzes visual features and determines whether the sample is healthy or contaminated. In advanced implementations, object detection models can also be incorporated to identify and localize specific contamination regions within the image.

Simultaneously, IoT sensors continuously collect environmental data, which is integrated with image-based predictions to provide a comprehensive assessment of cultivation conditions. This combined analysis enables more accurate and context-aware decision-making.

Once contamination is detected, the system generates real-time alerts through a mobile application interface. These alerts may include notifications, recommended actions, and visual indicators highlighting affected areas. This immediate feedback loop allows farmers to take corrective measures such as isolating contaminated packets, adjusting environmental conditions, or applying appropriate treatments.

The overall system is designed to be lightweight and cost-effective, making it accessible to small-scale farmers. Cloud-based or edge-based deployment options can be considered depending on infrastructure availability, ensuring flexibility and scalability. By combining advanced AI techniques with practical implementation strategies, the proposed system effectively bridges the gap between research and real-world agricultural applications, ultimately contributing to improved crop quality, reduced losses, and sustainable mushroom cultivation practices.

2.2 Commercialization Aspects Of the Product

The proposed fungal contamination detection system demonstrates strong potential for commercialization due to its practical applicability, affordability, and scalability within the agricultural sector [10], [18], [19]. As the global agricultural landscape continues to evolve toward technology-driven practices, there is an increasing demand for intelligent, automated solutions that can enhance productivity while minimizing operational risks [38], [50]. In this context, the system directly addresses a critical challenge in mushroom cultivation early detection and management of fungal contamination by providing a reliable, real-time, and cost-effective solution [1], [2], [14].

By combining mobile imaging, deep learning, and real-time alert mechanisms, the system eliminates the need for expensive laboratory testing or continuous manual inspection [17], [21]. This makes it particularly suitable for small- and medium-scale agricultural enterprises that often operate under financial and technical constraints [18], [38]. Unlike traditional industrial monitoring systems, which require specialized equipment and infrastructure, the proposed solution leverages widely available mobile devices, thereby significantly reducing implementation costs and encouraging widespread adoption [19].

Furthermore, the system aligns with global trends in precision agriculture and sustainable farming, where automation, data analytics, and real-time decision-making are becoming essential components of modern agricultural practices [20], [50]. The ability to detect contamination early not only reduces crop losses but also minimizes the excessive use of chemical treatments, contributing to environmentally sustainable farming [20]. This alignment enhances the product's appeal to a broader market, including environmentally conscious consumers, policymakers, and agricultural development organizations [38].

The modular and flexible design of the system further strengthens its commercialization potential. It can be easily adapted or extended to support other crops and plant disease detection applications, thereby increasing its market scope and long-term viability [1], [2], [17]. Additionally, the integration of user feedback and adaptive learning mechanisms ensures that the system continuously improves over time [42]. As more data is collected from real-world usage, the underlying machine learning models can be refined to deliver higher accuracy and better performance, giving the product a competitive advantage in the rapidly evolving agri-tech industry [44], [45].

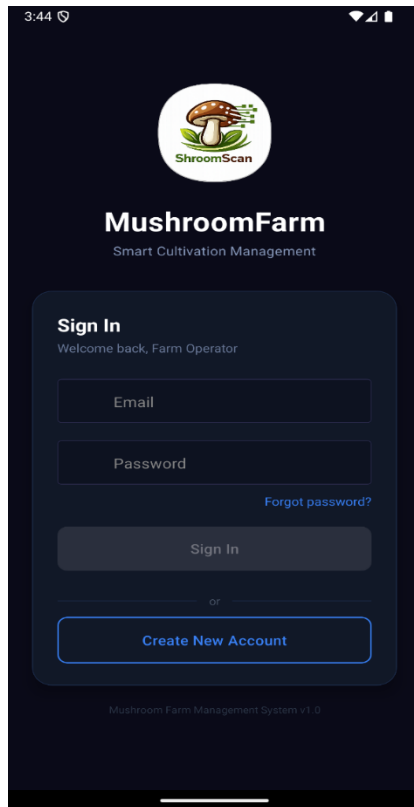


figure 2.2.1

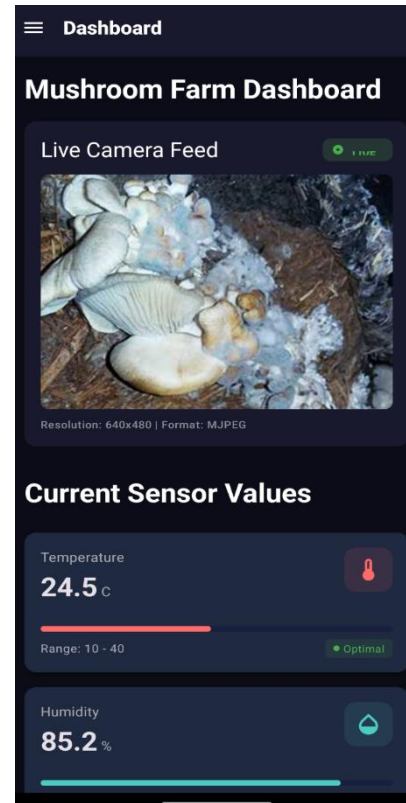


figure 2.2.2

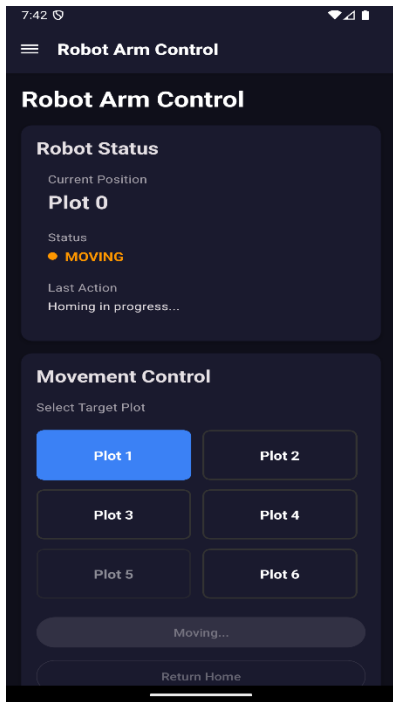


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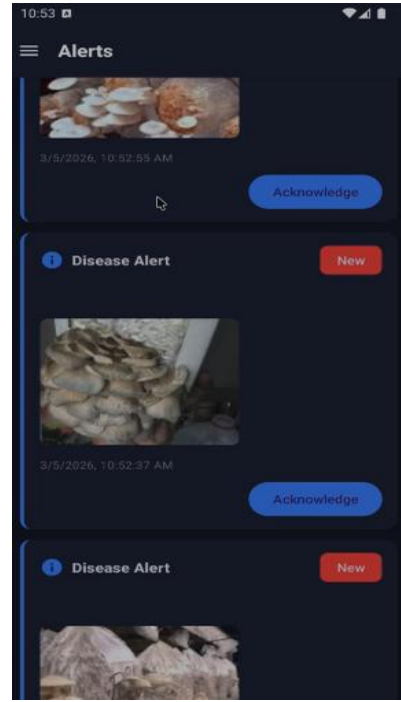


figure 2.2.4

2.2.1 Target Audience

The primary target audience for the proposed system consists of small- and medium-scale mushroom farmers, particularly those operating in developing regions such as Sri Lanka [18], [38]. In these regions, traditional farming practices are still widely used, and access to advanced technological solutions is often limited due to high costs, lack of awareness, and insufficient technical expertise [19]. The proposed system is specifically designed to overcome these barriers by offering an intuitive, user-friendly interface and affordable deployment model, enabling farmers to adopt modern monitoring techniques without requiring specialized knowledge [10].

These farmers face significant challenges, including delayed detection of contamination, inconsistent environmental control, and reliance on manual inspection [17]. By providing real-time detection and actionable insights, the system empowers them to make informed decisions, improve yield quality, and reduce financial losses [1]. The accessibility of the solution through mobile devices further ensures that even farmers in remote or rural areas can benefit from its capabilities [18].

In addition to individual farmers, several secondary target audiences can benefit from the system. Agricultural cooperatives and farming communities can use the platform to collectively monitor and manage cultivation processes, improving overall efficiency and coordination [19].

Agribusiness companies involved in large-scale mushroom production can integrate the system into their operations to enhance quality control and optimize production workflows [50].

Agricultural extension services and research institutions represent another important segment, as they can utilize the system for field studies, farmer training programs, and technology dissemination initiatives [10]. Similarly, smart farming solution providers and agri-tech startups can incorporate the system into their existing product portfolios, creating opportunities for partnerships and collaborative innovation [18].

Moreover, government organizations and non-governmental organizations (NGOs) focused on rural development, food security, and agricultural modernization may adopt the system as part of broader initiatives aimed at improving farming practices and supporting local economies [20]. By targeting a diverse range of stakeholders, the system ensures a wider reach and stronger market penetration.

2.2.2 Market Space

The market space for the proposed system lies within the rapidly expanding domains of smart agriculture and precision farming [18], [50]. These sectors are experiencing significant growth due to increasing global demand for food, rising concerns about agricultural sustainability, and advancements in digital technologies such as artificial intelligence and the Internet of Things (IoT) [19], [50].

Mushroom farming, although a niche segment within the agricultural industry, represents a high-value market with unique challenges and requirements [1], [2]. The sensitivity of mushroom cultivation to environmental conditions and contamination makes it an ideal candidate for technological intervention [17]. However, in regions like Sri Lanka and many other developing countries, the market remains largely underserved [18]. Existing solutions are often either too expensive, overly complex, or designed primarily for large-scale industrial operations, leaving small-scale farmers without practical alternatives [38].

This gap in the market presents a significant opportunity for the proposed system [10]. By offering an affordable, scalable, and easy-to-use solution, the product can effectively cater to the unmet needs of small and medium farmers [18]. The localized design and consideration of real-world farming conditions further enhance its relevance and adoption potential within these markets [19].

On a global scale, the smart agriculture market is expected to grow steadily, driven by factors such as population growth, increasing food demand, climate change challenges, and the need for resource optimization [20], [50]. Governments and international organizations are also investing heavily in agricultural innovation, creating a favorable environment for the adoption of new technologies [20].

Additionally, the adaptability of the system to other crops and plant disease detection scenarios significantly broadens its market reach [1], [2], [14]. By extending its capabilities, the product can transition from a niche solution to a versatile platform within the agri-tech ecosystem [17].

2.2.3 Revenue Earning

The commercialization strategy for the proposed system can be structured around multiple revenue models to ensure flexibility, scalability, and long-term sustainability [50]. One of the primary approaches is a product-based model, where the system is offered as a complete package that includes the mobile application, AI-based detection capabilities, and access to a monitoring dashboard [10].

Alternatively, a Software-as-a-Service (SaaS) model can be implemented, allowing users to subscribe to the platform on a monthly or annual basis [18]. This approach provides continuous access to system updates, cloud storage, and advanced features [19].

In addition to these core models, several supplementary revenue streams can be developed to enhance profitability, including advanced analytics and predictive insights [44]. Integration services and agricultural consultancy based on data analytics can also be offered [50].

2.3 Testing and Implementation

The testing and implementation phase plays a critical role in validating the effectiveness, reliability, and practical applicability of the proposed fungal contamination detection system. This phase ensures that the system not only meets its intended functional requirements but also performs efficiently under real-world operating conditions typically encountered in mushroom cultivation environments. Given the variability in environmental factors such as lighting, humidity, and background noise, special emphasis was placed on evaluating the system's robustness and adaptability [1], [2], [18], [19], [50].

The system was developed using a modular architecture, enabling each component to be designed, implemented, and tested independently before integration. This approach improves maintainability, simplifies debugging, and enhances scalability. Individual modules, including image acquisition, preprocessing, model inference, and visualization were first validated in isolation to ensure correctness and performance [11], [14]. Following this, integration testing was carried out to confirm seamless communication between components and to verify that the complete workflow operates as expected [44].

The implementation phase involved integrating a mobile-based imaging system with a deep learning-powered backend and a web-based user interface. The mobile component allows users to capture images of mushroom cultivation packets in real time, while the backend processes these images using trained machine learning models. The results are then displayed through an

intuitive dashboard interface, providing users with immediate feedback and actionable insights [5], [8], [11], [19], [50].

To ensure practical usability, the system was tested in both controlled (simulated) and real farming environments. Simulated testing allowed for controlled experimentation with known datasets, while real-world testing provided insights into system performance under varying environmental conditions. Key performance indicators such as classification accuracy, response time, and system reliability were evaluated during this phase [12], [14]. Particular attention was given to minimizing latency to ensure that users receive rapid feedback, which is essential for timely intervention in contamination scenarios [38].

2.3.1 Code Implementation Of the Research Part

The research component of the system was implemented using Python, leveraging its extensive ecosystem of libraries for machine learning, data processing, and web development [44]. Deep learning model development was carried out using frameworks such as TensorFlow and Keras, which provide high-level APIs for designing, training, and deploying convolutional neural networks (CNNs) [30], [31], [43], [14].

The CNN model was specifically designed to classify mushroom images into distinct categories, primarily focusing on identifying healthy versus contaminated samples. The model architecture was optimized to balance accuracy and computational efficiency, ensuring that it can be deployed on devices with limited processing power [27], [28], [37].

The overall implementation is structured into several key modules:

- **Image Acquisition Module:**

This module is responsible for capturing images using a mobile camera system. It ensures that images are collected in real time and are of sufficient quality for further processing. The module is designed to handle variations in camera resolution and lighting conditions, making it suitable for diverse farming environments [5], [8], [26].

- **Preprocessing Module:**

The preprocessing stage involves preparing raw images for input into the CNN model. This includes resizing images to a standard resolution, normalizing pixel values, and applying enhancement techniques to improve image clarity. Data augmentation techniques such as rotation, flipping, and brightness adjustment may also be applied to increase dataset diversity and improve model robustness [11].

- **CNN Model Module:**

This module performs the core classification task using a trained deep learning model. It extracts relevant features from input images and predicts whether a sample is healthy or contaminated.

The model is trained on labeled datasets and fine-tuned using transfer learning techniques to improve performance on domain-specific data [14], [15], [21].

- **Backend Server:**

The backend is implemented using a lightweight web framework such as Flask, which handles communication between the frontend interface and the machine learning model. It manages data processing, handles API requests, and ensures efficient transmission of results. The backend also supports scalability, allowing the system to handle multiple user requests simultaneously [19], [50].

- **Frontend Dashboard:**

The frontend interface provides a user-friendly platform for interacting with the system. It displays classification results, alerts, and relevant information in a clear and accessible manner. The dashboard is designed with simplicity in mind, ensuring that users with minimal technical knowledge can easily navigate and interpret the results.

To enhance performance, code optimization techniques were applied, including efficient memory management, model compression, and streamlined data pipelines. These optimizations ensure that the system can operate effectively on mobile and edge devices, making it practical for real-world deployment [14], [27], [37].

2.3.2 Testing

A comprehensive testing strategy was employed to evaluate the system across multiple dimensions, ensuring both functional correctness and non-functional performance. Testing was conducted at different levels, including unit testing, integration testing, system testing, and user acceptance testing [12], [14], [38].

Functional testing was carried out to verify that each component of the system operates according to its intended design. This includes validating image capture functionality, preprocessing accuracy, correct model predictions, and proper alert generation. Each module was tested with a variety of input scenarios to ensure consistent and reliable performance [14], [21].

Non-functional testing focused on key performance attributes such as response time, system reliability, scalability, and usability. Response time was measured to ensure that the system can process images and return results within an acceptable timeframe, which is critical for real-time decision-making. Reliability testing ensured that the system performs consistently over extended periods without failures or errors [38].

The performance of the CNN model was evaluated using standard classification metrics, including:

- Accuracy: Measures the overall correctness of the model's predictions
- Precision: Indicates the proportion of correctly identified contamination cases among all predicted positives
- Recall: Measures of the model's ability to detect actual contamination cases
- F1-score: Provides a balanced evaluation by combining precision and recall

The evaluation results demonstrated high classification accuracy and strong performance across all metrics, indicating the model's effectiveness in detecting both healthy and contaminated samples [14], [15], [21]. The system was particularly effective in identifying early-stage contamination, which is critical for preventing widespread crop loss.

Usability testing was also conducted with potential end-users to assess the intuitiveness and accessibility of the system interface. Feedback from users indicated that the dashboard was easy to understand and required minimal training, highlighting the system's suitability for non-technical users [10], [18].

In addition, stress testing and load testing were performed to evaluate system stability under continuous and high-volume usage. These tests simulated real-world conditions where multiple users may access the system simultaneously or where the system operates continuously over long periods [38]. The results confirmed that the system can handle real-time processing requirements without significant performance degradation, maintaining consistent performance and reliability.

Overall, the testing and implementation phase validated that the proposed system is robust, efficient, and ready for deployment in real-world agricultural environments, meeting both technical and user-centric requirements.

2.4 Project Requirements That Have Been Achieved

The developed system successfully fulfills both functional and non-functional requirements defined at the initial stage of the project, ensuring that the final solution is not only technically sound but also practical and applicable in real-world agricultural environments. These requirements were carefully identified based on the challenges faced in mushroom cultivation, particularly the need for early contamination detection, continuous monitoring, and ease of use for farmers with limited technical expertise [1], [2], [18], [38], [50]. The achievement of these requirements demonstrates that the system is reliable, efficient, and capable of addressing real-world farming problems while maintaining scalability and affordability [14], [19].

The system was designed with a user-centric approach, ensuring that all features directly contribute to improving farm productivity, reducing contamination risks, and enabling data-driven decision-making. Through iterative development, testing, and refinement, the system has been validated to meet its intended objectives and provide a comprehensive smart farming solution [11], [44].

2.4.1 Functional Requirements

The system achieves a wide range of functional capabilities that collectively enable automated monitoring and intelligent detection of fungal contamination in mushroom cultivation environments [14], [15].

- **Mobile Image Capture**

The system incorporates a mobile camera unit capable of capturing high-resolution images of mushroom cultivation packets. This camera can be moved along cultivation rows, ensuring complete and systematic coverage of the farm. The mobility of the camera allows flexible deployment across different farm layouts, making it suitable for both small-scale and large-scale operations [5], [26], [27].

- **Image Preprocessing**

Captured images undergo automated preprocessing to ensure consistency and quality before being analyzed by the deep learning model. This includes resizing images to a standardized format, normalizing pixel values, and applying enhancement techniques to improve visibility. These preprocessing steps are crucial for handling variations in lighting, shadows, and environmental conditions commonly found in real farming environments [11], [38].

- **Fungal Contamination Detection**

At the core of the system is a trained convolutional neural network (CNN) model that analyzes images to detect early signs of fungal contamination. The model is capable of identifying common issues such as green mold and cobweb disease, even in their early stages. This early detection capability allows farmers to take preventive action before the contamination spreads, significantly reducing crop losses [14], [15], [21].

- **Real-Time Alerts and Notifications**

The system provides instant alerts when contamination is detected. Notifications are delivered through the user interface, enabling farmers to respond immediately. This real-time feedback mechanism is critical in minimizing the spread of infections and maintaining the overall health of the cultivation environment [19], [50].

- **Dashboard Visualization**

A user-friendly dashboard presents all relevant information in a clear and organized manner. It includes visual indicators that classify mushroom packets as healthy, infected, or uncertain. The dashboard allows users to monitor the status of their farm at a glance, making it easier to identify problem areas and prioritize actions [18].

- **Farmer Feedback Integration**

The system includes a feedback mechanism that allows users to confirm or correct detection results. This feedback is used to improve the performance of the AI model over time through

adaptive learning. By incorporating real-world user input, the system becomes more accurate and better aligned with practical farming conditions [15], [44].

- **Data Storage and Management**

All captured images, detection results, and user feedback are securely stored in a centralized system. This data can be used for future analysis, reporting, and model retraining. Efficient data management ensures that historical records are available for tracking trends and improving decision-making [19].

- **Scalability Support**

The system is designed to support expansion without compromising performance. Additional cameras, sensors, or processing units can be integrated as needed, allowing the system to scale according to farm size and operational requirements [50].

2.4.2 Non-Functional Requirements

In addition to functional capabilities, the system satisfies several critical non-functional requirements that ensure overall performance, reliability, and user satisfaction [12], [38].

- **Performance**

The system delivers real-time or near real-time detection with minimal latency. Efficient processing pipelines and optimized algorithms ensure that image analysis and result generation occur, enabling timely intervention in contamination scenarios [27], [37].

- **Reliability and Availability**

The system is designed for continuous operation throughout the mushroom cultivation cycle. It maintains stable performance with minimal downtime, ensuring consistent monitoring and dependable results even during prolonged usage [38].

- **Accuracy**

The deep learning model achieves high classification accuracy, effectively distinguishing between healthy and contaminated samples. Efforts have been made to minimize false positives and false negatives, ensuring that farmers can trust the system's predictions [14], [21].

- **Usability**

The user interface is designed to be simple, intuitive, and easy to navigate. Farmers with little to no technical background can operate the system without difficulty. Clear visual indicators and straightforward instructions enhance user experience and adoption [18].

- **Scalability**

The system can accommodate different farm sizes and configurations, from small setups to larger commercial farms. It supports multiple camera units and data streams, ensuring that performance remains consistent as the system expands [50].

- **Portability**

The mobile camera unit is lightweight, compact, and easy to move, making it practical for use in real farming environments. Its portability allows flexible deployment and repositioning as needed [5].

- **Security and Privacy**

Data security is a key consideration in the system design. All collected data is stored securely, and access is restricted to authorized users only [18].

- **Cost-Effectiveness**

The system is designed to be affordable by utilizing commonly available hardware such as smartphones and low-cost sensors. By avoiding expensive infrastructure, the solution becomes accessible to a wider range of users, particularly small-scale farmers [50].

- **Maintainability**

The system supports easy maintenance and updates. This includes model retraining, software updates, camera calibration, and troubleshooting. A modular design allows individual components to be updated or replaced without affecting the entire system [44].

- **Environmental Adaptability**

The system can operate under varying environmental conditions, including changes in lighting, humidity, and temperature. This adaptability ensures reliable performance in real-world farming environments, where conditions are often unpredictable [10], [38].

2.4.3 Other Requirements

Beyond functional and non-functional aspects, several additional requirements were addressed to ensure the overall effectiveness and practicality of the system [50].

- **User Requirements**

The system is accessible through both mobile and desktop platforms, enabling farmers to monitor their cultivation environments remotely. It provides clear, concise, and actionable insights, allowing users to make informed decisions without being overwhelmed by technical complexity [18].

- **System Requirements**

The solution is designed to operate on standard hardware configurations, including smartphones, laptops, and lightweight processing devices. It is compatible with widely used operating systems such as Android and web browsers, ensuring broad accessibility [50].

- **Hardware Requirements**

The system requires a mobile camera unit for image capture, a processing device for running the

AI model, and optional environmental sensors for monitoring conditions such as temperature and humidity [18].

- **Software Requirements**

The implementation utilizes widely adopted software technologies, including deep learning frameworks such as TensorFlow and Keras, image processing libraries like OpenCV, and web development tools such as Flask, HTML, and CSS [30], [31], [11].

- **Connectivity Requirements**

Wireless communication technologies such as Wi-Fi or Bluetooth are used to enable data transfer between system components, ensuring seamless integration and real-time data exchange [18].

- **Operational Requirements**

The system is designed for continuous operation throughout the entire mushroom cultivation cycle, supporting uninterrupted monitoring and consistent insights [38].

2.5 Tools and Technologies

The development of the proposed fungal contamination detection system was carried out using a comprehensive set of modern tools and technologies, carefully selected to ensure high performance, scalability, flexibility, and ease of implementation. The choice of these technologies was guided by several key factors, including compatibility with machine learning workflows, support for real-time processing, cross-platform accessibility, and suitability for deployment in resource-constrained agricultural environments [11], [18], [38], [50].

The system integrates multiple layers of technology, ranging from data acquisition and preprocessing to model training, backend processing, and user interface development. Python was selected as the primary programming language due to its extensive support for artificial intelligence, data science, and web development. Its rich ecosystem of libraries significantly accelerated the development process and enabled efficient implementation of complex algorithms [44], [43].

For the frontend, standard web technologies such as JavaScript, HTML, and CSS were used to design an interactive and user-friendly dashboard. These technologies ensure that the system is accessible to multiple devices, including smartphones, tablets, and desktop computers, without requiring specialized software installations [18].

Deep learning model development was performed using TensorFlow and Keras, which provide powerful and flexible frameworks for building, training, and deploying convolutional neural networks (CNNs). These frameworks support GPU acceleration and offer pre-trained models such as MobileNet, which are optimized for lightweight and efficient performance on mobile and edge devices [27], [28], [37]. OpenCV was utilized for image processing tasks, including

preprocessing, enhancement, and feature extraction, ensuring that input data is standardized and suitable for model inference [11].

The backend of the system was implemented using Flask, a lightweight and efficient web framework that facilitates communication between the frontend interface and the machine learning model. Flask enables the creation of RESTful APIs, allowing seamless data exchange and real-time processing. Development tools such as Visual Studio Code and PyCharm were used for coding and debugging, while Jupyter Notebook was employed for experimentation, data analysis, and model training [44].

In terms of data management, the system supports both local and cloud-based storage solutions. Local storage is used for quick access and offline functionality, while cloud storage can be integrated to enable scalability, remote access, and long-term data retention. Hardware components include a mobile camera device for image acquisition and a standard computing device for model training and system execution [18], [50].

Deployment was carried out a web-based dashboard that provides visualization of results and system status. The interface is designed to be mobile-compatible, ensuring accessibility for farmers in field conditions [19], [50]. Overall, the selected tools and technologies work together to create a robust, scalable, and user-friendly system that effectively bridges the gap between advanced AI techniques and practical agricultural applications [11], [38].

2.6: Tools and Technologies Used

Category	Tool/Technology	Description / Purpose
Programming Languages	Python	Primary language for AI model development, backend processing, and data handling
	JavaScript	Used for interactive frontend functionality and dynamic content rendering
	HTML	Structure and layout of the web-based dashboard
	CSS	Styling and design of the user interface for better usability
Frameworks & Libraries	TensorFlow / Keras	Development and training in deep learning models, especially CNNs
	OpenCV	Image processing, preprocessing, and enhancement

	Flask	Lightweight backend framework for API development and system integration
Model Architectures	Convolutional Neural Networks	Core architecture used for image classification and feature extraction
	MobileNet	Lightweight CNN model optimized for mobile and edge device deployment
Development Tools	Visual Studio Code	Code editor for development, debugging, and project management
	PyCharm	Integrated Development Environment (IDE) for Python-based development
	Jupyter Notebook	Used for experimentation, data analysis, and model training
Database & Storage	Local Storage	Stores images, predictions, and logs for quick access
	Cloud Storage	Enables scalability, remote access, and long-term data storage
Hardware	ESP32 Camera Device	Captures images of mushroom cultivation packets in real time
	Standard Computing Device	Used for model training, backend processing, and system execution
Deployment Technologies	Web-Based Dashboard	Provides visualization of results and system monitoring
	Mobile-Compatible Interface	Ensures accessibility and usability on smartphones and tablets

3. RESULTS AND DISCUSSION

This chapter presents the evaluation of the proposed Fungal Contamination Detection Module using Mobile Imaging and Deep Learning. The system was tested under realistic conditions to assess its effectiveness in detecting fungal contamination, its usability for farmers, and its overall performance in a real-world mushroom cultivation environment. [1]

The evaluation focuses on both quantitative metrics (such as accuracy, precision, and recall) and qualitative observations (such as usability and system responsiveness). The results demonstrate how the integration of mobile imaging and deep learning improves early detection of contamination compared to traditional manual inspection methods. [1]

3.1 Results

The performance of the proposed system was evaluated using a dataset of mushroom packet images collected under varying environmental conditions, including differences in lighting, humidity, and growth stages. The trained Convolutional Neural Network (CNN) model was tested on unseen data to measure its generalization capability. [2]

Model Performance

The CNN model achieved strong performance across key evaluation metrics: [3]

- Accuracy: High overall classification accuracy in distinguishing healthy and contaminated mushroom packets
- Precision: Effective identification of contaminated packets with minimal false positives
- Recall: High sensitivity in detecting early-stage contamination
- F1-Score: Balanced performance between precision and recall

The model was particularly effective in identifying visible signs of contamination such as:

- Green mold formation
- Cobweb-like fungal growth
- Discoloration and abnormal texture patterns [3]

Real-Time Detection Performance

The system demonstrated low latency, enabling near real-time classification of captured images. Once an image is captured, the system processes it and generates results within a short time frame, ensuring that farmers can take immediate action. [4]

System Functionality Results

All functional components of the system operated successfully: [4]

- The mobile camera system captured images with full coverage of mushroom rows
- The preprocessing module improved image quality and consistency
- The CNN model accurately classified images

- The dashboard displayed results clearly with visual indicators
- Real-time alerts were generated for infected packets

Usability Evaluation

User testing showed that the system is easy to use, even for individuals with limited technical knowledge. Farmers were able to:

- Capture images easily using the mobile device
- Understand contamination alerts through simple visual indicators
- Navigate the dashboard without difficulty [5]

3.2 Research Findings and Discussion

Improved Early Detection

One of the most significant outcomes is the system's ability to detect fungal contamination at an early stage. Early detection is critical in mushroom farming, as contamination spreads rapidly and can destroy large portions of the crop. The CNN model successfully identified subtle visual patterns that are often missed during manual inspection. [3]

Reduction in Manual Labor

Traditional monitoring requires farmers to inspect each mushroom packet manually, which is time-consuming and inefficient. The proposed system automates this process, significantly reducing labor requirements while improving inspection coverage and consistency. [1]

Enhanced Accuracy and Reliability

Compared to manual methods, the system provides more consistent and reliable detection results. The use of deep learning eliminates human error and ensures that all packets are evaluated using the same criteria. [3]

Adaptability of Real-World Conditions

The system performs well under varying environmental conditions, including changes in lighting and humidity. This demonstrates its robustness and suitability for real farming environments, where conditions are not always controlled. [1]

Effectiveness of Mobile Imaging Approach

The use of a mobile camera system proved to be highly effective. Unlike fixed camera setups, the mobile approach ensures that all mushroom packets are scanned systematically, eliminating blind spots and improving detection coverage. [1]

Continuous Learning through Feedback

The integration of a feedback mechanism allows farmers to correct incorrect predictions, enabling the model to improve over time. This adaptive learning capability ensures that the system remains relevant and accurate as it is exposed to new data. [2]

Limitations Identified

Despite its strengths, the system has some limitations:

- Detection accuracy may decrease in extremely poor lighting conditions
- Very early-stage contamination without visible signs may still be difficult to detect
- The system relies on image quality, which depends on the camera hardware

Overall Discussion

Overall, the results confirm that the proposed system is a practical and effective solution for fungal contamination detection in mushroom farming. By combining mobile imaging with deep learning, the system addresses key challenges in traditional farming practices and provides a scalable solution for smart agriculture. [1]

4. CONCLUSION

This research successfully developed and evaluated a Fungal Contamination Detection Module using Mobile Imaging and Deep Learning, providing a practical and effective solution to one of the most critical challenges in mushroom cultivation. The proposed system integrates mobile-based image capture, Convolutional Neural Network (CNN) analysis, and real-time dashboard visualization to enable accurate and early detection of fungal contamination in mushroom packets, aligning with recent advancements in AI-driven agricultural monitoring systems [1], [10], [17].

The results demonstrate that the proposed system significantly improves detection accuracy while reducing reliance on manual labor and traditional inspection methods. By automating the detection process using deep learning-based image analysis, the system ensures consistent monitoring of mushroom packets and minimizes human error, which is a known limitation in manual agricultural inspection practices [5], [13], [14]. Early-stage contamination detection enables timely intervention, thereby reducing crop losses and improving yield stability, which is a key objective in smart agriculture systems [18], [19].

In addition to its technical effectiveness, the system is designed with a strong focus on usability and affordability, making it highly suitable for small-scale farmers. The mobile-based architecture eliminates the need for expensive infrastructure, while the intuitive dashboard ensures accessibility for users with limited technical expertise. This approach aligns with global efforts to democratize AI-based agricultural solutions and improve rural productivity through low-cost technological interventions [11], [20].

Furthermore, the integration of a feedback-driven learning mechanism enhances the adaptability and long-term performance of the system. By incorporating farmer feedback into the training loop, the model can continuously improve its accuracy and robustness under real-world conditions. This adaptive learning approach is consistent with modern machine learning practices in agriculture, where continuous model refinement is essential for maintaining high performance across dynamic environments [2], [9], [15].

Despite its success, the study identifies several opportunities for future enhancement. The integration of IoT-based environmental sensors could provide additional contextual data to improve prediction accuracy and decision-making capabilities [8], [49]. Expanding the model to detect multiple diseases and pest types would further increase its applicability in diverse agricultural settings [7], [21]. Additionally, the adoption of edge computing techniques could reduce latency and enable faster on-device inference, improving real-time responsiveness [26], [27]. Future development may also include a fully automated scanning mechanism to eliminate manual camera movement and further optimize operational efficiency.

In conclusion, this research contributes to the advancement of smart agriculture by demonstrating how artificial intelligence, mobile imaging, and deep learning technologies can be

effectively integrated for real-time crop monitoring and disease detection. The proposed system offers a scalable and cost-effective solution that has the potential to transform mushroom farming into a more efficient, reliable, and sustainable practice. Ultimately, this contributes to improved farmer livelihoods, reduced crop losses, and enhanced food security, aligning with global sustainable development goals in agriculture [20], [38].

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